

LLM Recommendation Report: Complaint Classification and Legal Document Retrieval

This report analyzes the six provided LLMs (KoBERT, klue/RoBERTa-large, BGE-m3, KoSimCSE, Solar 10.7B, EXAONE-3.0-7.8B) to recommend the two most suitable models for two core tasks: **Complaint Classification** and **General Legal Document Retrieval**.

1. Optimal Model for Complaint Classification: klue/RoBERTa-large

For accurately classifying complaint categories or assigning them to the appropriate departments, **klue/RoBERTa-large** is highly recommended.

- **Sophisticated Korean NLU:** Trained on a massive Korean corpus (including news, web documents, and national petitions), it excels at understanding the complex mix of administrative terminology and colloquialisms often found in civil complaints.
- **Balance of Efficiency and Performance:** With approximately 337M parameters, it is much lighter and faster than generative LLMs. It achieves outstanding classification accuracy when fine-tuned on small amounts of complaint data.
- **Superiority over KoBERT:** Compared to its encoder counterpart KoBERT, it guarantees higher performance due to superior training data volume and quality, as well as model size.

2. Optimal Model for Legal Document Retrieval: BGE-m3

For searching through extensive legal articles, precedents, and lengthy documents, **BGE-m3** is the ideal choice.

- **Long Context Support:** Unlike standard embedding models limited to 512 tokens, BGE-m3 can process up to 8,192 tokens. This allows for intact embedding and retrieval of lengthy legal documents without truncation.
- **Hybrid Search Capability:** It simultaneously supports dense retrieval (semantic search) and sparse retrieval (keyword matching/BM25), maximizing accuracy for legal searches where exact terminology is crucial.

 Note: Korean Language Support in BGE-m3

While BGE-m3 supports over 100 languages, **Korean is definitively and robustly included**. As a state-of-the-art multilingual embedding model, its Korean retrieval accuracy is exceptional. It is currently widely adopted as the standard for building Korean RAG (Retrieval-Augmented Generation) systems and internal enterprise search engines.

3. Comparison Summary of the 6 Candidate Models

Model Name	Primary Purpose	Context Length	Evaluation & Recommendation
klue/RoBERTa-large	Text Classification/NLU	512 tokens	[Selected] Optimal for complaint classification.
BGE-m3	Retrieval/Embedding	8,192 tokens	[Selected] Optimal for legal document search/RAG.
KoBERT	Text Classification	512 tokens	Lightweight, but inferior to RoBERTa-large.
KoSimCSE	Sentence Similarity	512 tokens	For short sentences; unsuitable for long documents.
Solar 10.7B	Text Generation	4,096 tokens	Generative LLM; excessive resource consumption for simple classification.
EXAONE-3.0-7.8B	Text Generation	8,192 tokens	Generative LLM; suitable for generating final answers post-retrieval.

Recommended Pipeline Architecture:

We strongly recommend a 2-track hybrid architecture: utilize **klue/RoBERTa-large** to rapidly classify incoming complaints, and leverage **BGE-m3** to search an internal Vector DB for the legal grounds necessary to resolve and answer the queries.